Lecture 6

Composite systems in infinite dimension: EPR

Objectives. *A part of an entangled pure state can have a mixed state even with infinitely many levels. We construct the composite space and its Schmidt form, then define the partial trace by the observations made on one part. EPR's position–momentum correlations motivate a quantitative test: a separable preparation has a lower bound on two collective variances.*

- Prove the Schmidt decomposition for a unit vector of two separable Hilbert spaces and identify its reduced states.
- Construct the partial trace from bounded expectations, including mixed composite states.
- State EPR's premises and prove a separability bound; evaluate it on the specified squeezed covariances without treating a delta function as a state.

6.1 The composite as square-summable arrays

QM-I used a finite tensor product: coefficients c_{jk} specified a vector $\sum_{jk} c_{jk} e_j \otimes f_k$. Now let $\mathcal{H}_1, \mathcal{H}_2$ be nonzero separable complex Hilbert spaces, with fixed orthonormal bases (e_j), (f_k). Their *Hilbert tensor product* is the space of arrays with

$$\Psi = \sum_{j,k} c_{jk} e_j \otimes f_k, \qquad \|\Psi\|^2 = \sum_{j,k} |c_{jk}|^2 < \infty.$$

The inner product is the corresponding sum $\sum \bar{c}_{jk} d_{jk}$; Cauchy–Schwarz makes it convergent. Finite arrays are dense, by truncating the nonnegative norm sum. The product $u \otimes v$ has coefficients $u_j v_k$ and norm $\|u\| \, \|v\|$. For any other ONBs (u_ℓ), (v_m), their products form an ONB in this same space. Orthonormality follows from $\langle u \otimes v, u' \otimes v' \rangle = \langle u, u' \rangle \, \langle v, v' \rangle$, first on finite sums and then by Parseval. For completeness, if Ψ is orthogonal to every $u_\ell \otimes v_m$, fix ℓ. Its coefficient row $(\langle u_\ell \otimes f_k, \Psi \rangle)_k$ is orthogonal to every v_m, so is zero. Each old column is then orthogonal to every u_ℓ, and also zero. Thus $\Psi = 0$. The unitary identification between array descriptions sends each product $u \otimes v$ to that same product and preserves all inner products; its inverse is the reverse basis change.

For $L^2(\mathbb{R})$ factors, the product basis identifies this tensor product with $L^2(\mathbb{R}^2)$, by standing fact S3. Product functions become $(u \otimes v)(x_1, x_2) = u(x_1) v(x_2)$; general vectors need not factor. For a bounded A on part 1, define $A \otimes \mathbb{1}$ first on finite sums. Grouping coefficients by the second index gives

$$\|(A \otimes \mathbb{1})\Psi\|^2 = \sum_k \left\| A\Big(\sum_j c_{jk} e_j \Big) \right\|^2 \le \|A\|^2 \, \|\Psi\|^2 .$$

Thus it extends to the whole tensor product. Unit product vectors show its norm is exactly $\|A\|$. Its adjoint is $A^\dagger \otimes \mathbb{1}$ by testing finite tensors and continuity. If $A \geq 0$, S6 writes it as $(\sqrt{A})^\dagger \sqrt{A}$, so $A \otimes \mathbb{1} \geq 0$. These bounded operators represent all observations confined to part 1.

Self-check 6.1

1. Why is square summability exactly the condition for the array to define a vector?
2. Can a bounded local observation have a larger operator norm on the composite?

6.2 Schmidt's theorem without a finite-rank assumption

Theorem 6.1 (Schmidt decomposition). *Every unit $\Psi \in \mathcal{H}_1 \otimes \mathcal{H}_2$ has a finite or countable expansion*

$$\Psi = \sum_\ell \sqrt{p_\ell}\, u_\ell \otimes v_\ell, \quad p_\ell > 0, \quad \sum_\ell p_\ell = 1,$$

where the two displayed families are orthonormal. The sum converges in norm. Zero coefficients can be omitted; neither family need be a basis of its whole factor.

Proof. For the coefficient array above, form a *linear* operator $C : \mathcal{H}_2 \to \mathcal{H}_1$ by $(Cx)_j = \sum_k c_{jk} x_k$. Row-wise Cauchy–Schwarz gives $\|Cx\|^2 \leq \sum_{jk} |c_{jk}|^2 \, \|x\|^2$. Put $R = CC^\dagger$, the analogue of QM-I's coefficient formula for the first reduced state. It is bounded and positive. Its diagonal sum in (e_j) is

$$\sum_j \langle e_j, R e_j \rangle = \sum_j \left\| C^\dagger e_j \right\|^2 = \sum_{j,k} |c_{jk}|^2 = 1.$$

Proposition 1.4 gives compactness and an orthonormal eigenbasis, including the kernel; its nonzero eigenvalues are $p_\ell > 0$ with $\sum_\ell p_\ell = 1$. Let u_ℓ be their orthonormal eigenvectors. The coefficient of $u_\ell \otimes f_k$ in Ψ is $d_{\ell k}$:

$$d_{\ell k} = \langle u_\ell \otimes f_k, \Psi \rangle .$$

Thus define

$$d_{\ell k} = \sum_j \overline{(u_\ell)_j} c_{jk}, \qquad v_\ell = \frac{1}{\sqrt{p_\ell}} \sum_k d_{\ell k} f_k.$$

The components of $C^\dagger u_\ell$ are $\bar{d}_{\ell k}$, not $d_{\ell k}$. Consequently

$$\langle v_\ell, v_m \rangle = \frac{\langle C^\dagger u_m, C^\dagger u_\ell \rangle}{\sqrt{p_\ell p_m}} = \frac{\langle u_m, R u_\ell \rangle}{\sqrt{p_\ell p_m}} = \delta_{\ell m}.$$

For a basis vector u of $\ker R$, $\left\| C^\dagger u \right\|^2 = \langle u, Ru \rangle = 0$, so the corresponding coefficient row of Ψ is zero. Expand the first factor in this full eigenbasis of R. Its nonzero rows are precisely $\sqrt{p_\ell} v_\ell$, proving the stated tensor expansion. Parseval gives the squared tail $\sum_{\ell > N} p_\ell \to 0$, proving norm convergence as well as completeness of the expansion. ∎

The basis coefficient is conjugated because $\langle u_\ell, e_j \rangle = \overline{(u_\ell)_j}$. Thus C is an auxiliary linear matrix map, whereas the tensor coefficient formula contains $u_j v_k$, without conjugating either factor. The displayed component calculation keeps that convention explicit. Degenerate weights permit changes of Schmidt vectors within their eigenspaces; the resulting expansion need not be unique within degenerate subspaces.

For a product $c_{jk} = a_j b_k$, the range of C is one dimensional; this construction therefore gives exactly one nonzero Schmidt weight. Conversely a one-term Schmidt expansion is a product. More than one weight therefore detects pure-state entanglement, just as in QM-I; infinite Schmidt rank is allowed. For instance, for $0 < r < 1$ the vector

$$\Psi = \sqrt{1-r}\sum_{n=0}^{\infty} r^{n/2} e_n \otimes f_n$$

has weights $(1-r)r^n$. If terms $0, \dots, N$ are retained, its omitted norm is $r^{(N+1)/2}$. Each finite truncation can approximate the vector, although none has its exact infinite-rank reduced state.

Self-check 6.2

1. In the proof, why must the second factor use $d_{\ell k}$ rather than its complex conjugate?
2. What are the Schmidt weights of $(e_0 \otimes f_0 + i e_1 \otimes f_1)/\sqrt{2}$?

6.3 The state of a part

For a Schmidt vector, orthogonality of the v_ℓ gives, for bounded A,

$$\langle \Psi, (A \otimes \mathbb{1})\Psi \rangle = \sum_{\ell} p_\ell \, \langle u_\ell, A u_\ell \rangle \, .$$

One can obtain this by finite truncations: boundedness controls the norm limit and the expectation series is absolutely convergent. Thus $\rho_1 = \sum_\ell p_\ell \, |u_\ell\rangle\langle u_\ell|$ is the density operator of part 1; similarly $\rho_2 = \sum_\ell p_\ell \, |v_\ell\rangle\langle v_\ell|$. They have the same nonzero eigenvalues. Neither is generally pure even when the composite is. Every Schmidt expansion gives this bounded local expectation rule. L1's uniqueness therefore shows that its weights are the nonzero eigenvalues of ρ_1, including multiplicities, independent of the construction.

Theorem 6.2 (Partial trace). *For every density operator ρ on $\mathcal{H}_1 \otimes \mathcal{H}_2$, there is a unique density operator $\rho_1 = \mathrm{Tr}_2 \, \rho$ such that*

$$\mathrm{Tr}(\rho_1 A) = \mathrm{Tr}\left(\rho(A \otimes \mathbb{1})\right) \quad \textit{for every bounded } A \textit{ on } \mathcal{H}_1. \tag{6.1}$$

Proof. Define the right-hand side as a functional $\omega_1(A)$. It is linear, positive and normalised by L1 and the tensor properties above. To check normality, take orthogonal projections E_j on part 1 with strong sum E. Their tensor extensions have strong sum $E \otimes \mathbb{1}$: this is immediate on a product vector from the part-1 strong convergence, extends to finite sums, then to every vector by density and the uniform norm bound one for the partial projection sums. For any spectral density expansion $\rho = \sum_n p_n \, |\Psi_n\rangle\langle\Psi_n|$, it follows that

$$\omega_1(E) = \sum_n p_n \sum_j \left\langle \Psi_n, (E_j \otimes \mathbb{1})\Psi_n \right\rangle = \sum_j \omega_1(E_j).$$

All summands are nonnegative, so interchange is justified. L1's normal-state theorem supplies a unique ρ_1, which is exactly (6.1). ∎

This construction also proves affine dependence on ρ, by uniqueness. For mixtures, one may compute each pure reduced state and mix them with the same weights; the normal-state proof shows the result does not depend on a particular composite preparation. For density operators $\tau_1 = \sum_j a_j \, |u_j\rangle\langle u_j|$ and $\tau_2 = \sum_k b_k \, |v_k\rangle\langle v_k|$, define their product as $\tau_1 \otimes \tau_2 =$

$\sum_{jk} a_j b_k \,|u_j \otimes v_k\rangle\langle u_j \otimes v_k|$. It is positive with trace one, and bounded product expectations factor. Its partial trace returns τ_1. A purification also exists: from $\rho_1 = \sum_n p_n \,|u_n\rangle\langle u_n|$ use $\sum_n \sqrt{p_n} u_n \otimes f_n$ with a large enough second space. The squared norm sum is $\sum_n p_n = 1$, and the bounded first-factor expectations give ρ_1 by the preceding Schmidt calculation. QM-I, Theorem 11.10, uses the conjugated eigenvectors as partner family; replacing them by any orthonormal (f_n) leaves these expectations unchanged.

For the geometric Schmidt example, $\rho_1 = (1-r)\sum_n r^n \,|e_n\rangle\langle e_n|$. A local event $|e_0\rangle\langle e_0|$ has probability $1-r$, although the composite vector remains pure. Tracing a part is a restriction of observations, not a random selection of one term of a Schmidt expansion. Compare L5: restriction to a smaller algebra can produce a mixed state, without changing the initial vector by any dynamics.

Self-check 6.3

1. For the geometric Ψ, compute the partial trace of $\frac{1}{2}\,|\Psi\rangle\langle\Psi| + \frac{1}{2}\,|e_0 \otimes f_1\rangle\langle e_0 \otimes f_1|$.
2. Does the mixedness of ρ_1 imply that the composite was prepared as a mixture?

6.4 What EPR asked

Einstein, Podolsky and Rosen (1935) considered a pair with a fixed relative position and a fixed total momentum. Write $X_i = Q^{(i)}$ for position on part i in this section, and $P^{(i)}$ for momentum on part i. In the ideal description, reading X_1 would predict X_2 exactly; alternatively reading $P^{(1)}$ would predict $P^{(2)}$ exactly. The inference concerns the distant partner, not simultaneous sharp readings of X_1 and $P^{(1)}$ on the same particle.

Their criterion of reality says that certain prediction of a physical quantity, without disturbing the system, is sufficient for an element of reality. Their locality premise says that the choice of reading on part 1 does not change the physical reality of the separated part 2. They then assign both position and momentum reality to part 2, independent of which reading is chosen on part 1. Completeness would require the state description to represent all those elements of reality. A single wavefunction cannot give both local quantities sharp values. EPR therefore conclude that the wavefunction description is incomplete: otherwise the two inferred local quantities could not both be real. That argument depends on the reality and locality premises just stated; L13 examines locality more closely. Exact certainty also uses their ideal correlations; normalised preparations give finite inference errors.

The ideal expression $\delta(x_1 - x_2 - x_0)$ is not a normalised vector of $L^2(\mathbb{R}^2)$. L1's shrinking-bump test, applied to strips about $x_1 - x_2 = x_0$, excludes its exact position expectations; we give the strip estimate below. As in QM-II's plane wave, writing it formally does not create a state. Actual experiments have finite correlations. We can quantify one consequence without importing an ideal delta function into the state space.

On $\mathcal{S}(\mathbb{R}^2)$ direct differentiation gives

$$[X_1 - X_2, P^{(1)} + P^{(2)}] = [X_1, P^{(1)}] - [X_2, P^{(2)}] = 0.$$

There is therefore no local Heisenberg lower bound between these two collective quantities from their commutator. This does not remove the uncertainty relation for each local $X_i, P^{(i)}$. The next bound says what product preparations, and their mixtures, still cannot do.

Self-check 6.4

1. Which EPR premise connects a choice of experiment on part 1 to reality assigned to part 2?
2. Why is a delta expression not an admissible substitute for a density operator?

6.5 A variance bound for separable preparations

A *finite separable preparation* is $\rho = \sum_{a=1}^{m} w_a |u_a \otimes v_a\rangle\langle u_a \otimes v_a|$, with $w_a \geq 0$, $\sum_a w_a = 1$, unit $u_a, v_a \in \mathcal{S}(\mathbb{R})$. All position and momentum moments used below are finite. For each collective B define $\langle B\rangle_\rho = \sum_a w_a \langle u_a \otimes v_a, B(u_a \otimes v_a)\rangle$, and similarly for B^2; variance is the second moment minus the square of the mean. These depend only on ρ: its finite-dimensional range lies in Schwartz space, so $B\rho$ and $B^2\rho$ are finite-rank bounded operators, whose finite matrix traces give exactly those sums. A general separable state is usually defined using a suitable closed convex hull; our proof is for this stated finite class, with the finite-moment extension named rather than hidden in a continuity claim for unbounded observables.

Proposition 6.3 (Duan-type bound, finite preparation form)**.** *For the finite separable preparations above, and every length $s > 0$,*

$$\frac{\mathrm{Var}_\rho(X_1 - X_2)}{s^2} + \frac{s^2 \,\mathrm{Var}_\rho(P^{(1)} + P^{(2)})}{\hbar^2} \geq 2. \tag{6.2}$$

Proof. For a product vector, expectations factor, so the position covariance and momentum covariance between the parts vanish. Its left side is the sum of the two local contributions. If $\alpha_i = \Delta X_i / s$ and $\beta_i = s\Delta P^{(i)}/\hbar$, QM-II, Theorem 4.9, on Schwartz functions gives $\alpha_i\beta_i \geq 1/2$. Hence $\alpha_i^2 + \beta_i^2 \geq 2\alpha_i\beta_i \geq 1$, and summing the parts gives 2. For a finite mixture and either collective observable B, put $b_a = \langle B\rangle_a$, $\bar{b} = \sum_a w_a b_a$. Expanding its second moment gives the law of total variance

$$\mathrm{Var}_\rho B = \sum_a w_a \,\mathrm{Var}_a B + \sum_a w_a (b_a - \bar{b})^2 \geq \sum_a w_a \,\mathrm{Var}_a B.$$

Apply it to both collective observables and average the product bounds. ∎

The scale s converts position and momentum variances into dimensionless numbers. Changing it cannot allow a finite separable preparation to violate the inequality. The broader separable-state version with finite second moments is a standard continuous-variable entanglement criterion (cited: Duan et al., *PRL* **84**, 2722–2725 (2000)). Mere trace norm convergence of states does not justify continuity of unbounded second moments, which is why it is not used as a shortcut here.

Consider the normalised Gaussian family

$$\psi_r(x_1, x_2) \;\propto\; \exp\left[-\frac{e^{2r}(x_1 - x_2)^2}{4s^2} - \frac{e^{-2r}(x_1 + x_2)^2}{4s^2}\right], \quad r \geq 0.$$

For finite r this is a Schwartz vector. Both collective means vanish. Its covariances, supplied here for the test, are

$$\mathrm{Var}(X_1 - X_2) = s^2 e^{-2r}, \qquad \mathrm{Var}(P^{(1)} + P^{(2)}) = \frac{\hbar^2}{s^2} e^{-2r}.$$

Their Gaussian-integral derivation is a computation for Sheet 6; no general proof is deferred there. Inserting them gives $2e^{-2r}$, below 2 for every $r > 0$. Thus these vectors cannot be finite separable preparations; the cited broader criterion also certifies their entanglement among all separable states. The practical value of the test is that it uses only two second moments, including for noisy mixed preparations without a pure-state Schmidt form. For this pure example entanglement could also be checked by nonfactorisation. At $r = 0$ the exponent factors into independent Gaussians and the bound is saturated. At $r = (\log 2)/2$

the sum equals 1: the two terms are each $1/2$. The test concerns correlations, not either local variable becoming simultaneously sharp with its local conjugate.

The local uncertainties increase even though both tested collective variances decrease. To compute the local position variances, use

$$u = \frac{x_1 - x_2}{\sqrt{2}}, \qquad v = \frac{x_1 + x_2}{\sqrt{2}}.$$

The position density factors in these coordinates. The Gaussian variances are $s^2 e^{-2r}/2$ for u and $s^2 e^{2r}/2$ for v. The inverse relations are $x_1 = (u + v)/\sqrt{2}$ and $x_2 = (v - u)/\sqrt{2}$. Independence removes the covariance term, giving

$$\operatorname{Var} X_i = \frac{s^2}{4}(e^{-2r} + e^{2r}) = \frac{s^2}{2}\cosh(2r), \qquad i = 1, 2.$$

For a real Gaussian factor, QM-II's Gaussian calculation gives momentum variance $\hbar^2/(4 \operatorname{Var} u)$, with u replaced by v for the other factor. The momentum variances of the two Gaussian factors are $\hbar^2 e^{2r}/(2s^2)$ and $\hbar^2 e^{-2r}/(2s^2)$. The same inverse coordinate change therefore gives

$$\operatorname{Var} P^{(i)} = \frac{\hbar^2}{2s^2}\cosh(2r), \qquad \Delta X_i \Delta P^{(i)} = \frac{\hbar}{2}\cosh(2r) \geq \frac{\hbar}{2}.$$

A narrow relative position is compatible with a broad distribution for each particle separately. Figure 6.1 shows this distinction using position probabilities alone.

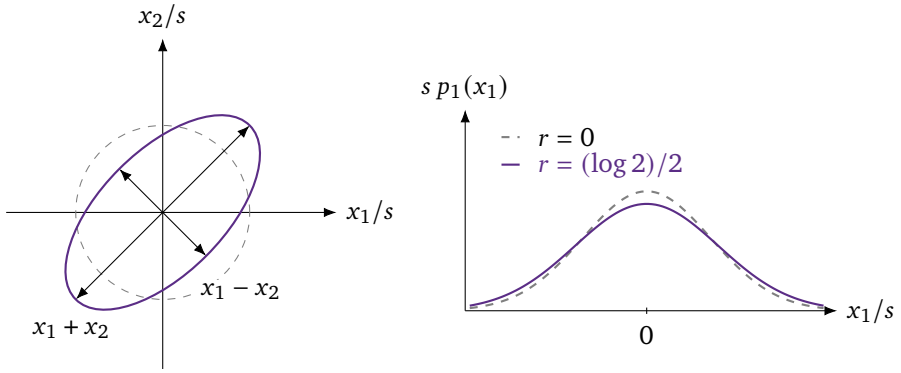


Figure 6.1: An equal-density contour and the normalised first-position marginal at $r = 0$ (dashed) and $r = (\log 2)/2$ (solid). Here p_1 is the first part's position density, and the vertical scale is $sp_1(x_1)$. The correlation narrows across the diagonal while the local marginal broadens; these are position plots, not joint position–momentum distributions.

As $r \to \infty$, the collective variances tend to zero, but the vectors have no norm limit. Here is the promised version of L1's bump test. Let f_N be continuous, $0 \leq f_N \leq 1$, equal to one at zero and zero outside $[-1/N, 1/N]$. For a smooth function v supported in $[-R, R]^2$,

$$\langle v, f_N(X_1 - X_2)v\rangle \leq \frac{4R}{N}\,\|v\|_\infty^2 \longrightarrow 0,$$

because the intersecting strip has width at most $2/N$ in each fixed x_2 slice. Approximation by such v (S4) and the multiplier bound one extend this limit to every L^2 vector. A finite spectral head and a small positive tail extend it to every density operator. Translating

the strip gives the same result about x_0. Thus no normal state has the exact expectations $\langle f(X_1 - X_2)\rangle = f(0)$ for every bounded continuous f, as a delta would require. For fixed f, the squeezed expectations do tend to $f(0)$: outside a strip of width ϵ the probability is at most $\mathrm{Var}(X_1 - X_2)/\epsilon^2$, by integrating $\epsilon^2 1_{|x_1-x_2|>\epsilon} \le (x_1 - x_2)^2$; inside it use continuity at zero. A hypothetical norm limit would preserve bounded expectations and give the excluded delta rule. Increasingly sharp correlations do not supply a limiting state in the Hilbert space.

The physics

Two-mode squeezed light realises analogous correlated quadratures. Ou, Pereira, Kimble and Peng, *PRL* **68**, 3663 (1992), reported an inferred-variance product 0.70 ± 0.01 below their unit threshold. That is Reid's EPR inference criterion, not the sum in (6.2). The optical experiment motivates the same position–momentum logic, but its quoted statistic must not be inserted into our sum bound. Our computed prediction is $2e^{-2r}$ for the explicitly specified ideal covariances; loss and detector noise require their own model.

Self-check 6.5

1. Where does mixing increase, rather than decrease, a variance in the proof?
2. Evaluate the sum for $r = \log 2$. Does it certify the original EPR conclusion about completeness by itself?

What is a composite, and what is a part?

Without superselection, a quantum system has a separable Hilbert space, its normal states and its bounded observables. The composite space is the Hilbert tensor product. The state of a part is the unique partial trace reproducing its bounded local expectations. Entanglement can make that state mixed even when the composite is pure. Superselection requires the specified observable algebra of L5 as well.

Summary. Square-summable arrays extend QM-I's composite rule. Compactness and S5 give Schmidt's theorem; L1 gives the mixed-state partial trace from local expectations. EPR adds premises about reality, locality and completeness. Normalised squeezed states replace its ideal delta expression by finite correlations, and their variance sum violates the product-mixture bound. Neither a reduced mixture nor a limiting delta selects a definite global alternative.

Exercises. Try Sheet 6 in the separate *Exercise sheets* booklet (English).

Self-check answers.

6.1 1. The basis is orthonormal, so the squared norm of every tail is the sum of its coefficient moduli squared. 2. No; its norm equals the original norm, with equality tested on products.

6.2 1. The expansion coefficient is $\langle u_\ell, e_j\rangle = \overline{(u_\ell)_j}$; the tensor coefficient itself is $u_j v_k$. 2. $1/2, 1/2$; absorb i into the second Schmidt vector.

6.3 1. $\frac{1}{2}\rho_1 + \frac{1}{2}\,|e_0\rangle\langle e_0|$. 2. No; a pure entangled composite is a counterexample.

6.4 1. Locality, combined with the criterion of reality for certain prediction. 2. It would give delta expectations for every bounded continuous function of the relative position, excluded by the strip-bump test.

6.5 1. The nonnegative variance of the component means. 2. $2e^{-2\log 2} = 1/2$; no, the entanglement test alone does not establish all EPR premises.

Sources. Hall, *Quantum Theory for Mathematicians*, Theorem 19.13 pp. 430–431; Moretti, *Spectral Theory and Quantum Mechanics*, §13.4 pp. 844–858. The matrix compactness and normal-state proofs above supply the needed prerequisites explicitly. Einstein–Podolsky–Rosen, *Phys. Rev.* **47**, 777 (1935); Duan et al., *PRL* **84**, 2722 (2000). The optical experiment and its reference are in the full edition.